

Talking Wikidata: Communication patterns and their impact on community engagement in collaborative knowledge graphs

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ABSTRACT

We study collaboration patterns of Wikidata, one of the world's largest collaborative knowledge graph communities. Wikidata lacks long-term engagement with a small group of priceless members, 0.8%, to be responsible for 80% of contributions. It is thus essential to investigate behavioural patterns to enhance members' contributions and participation. Previous work focused on community interactions using activity logs and semi-structured interviews to investigate Wikidata contributors' engagement. However, no prior work has explored members' communication patterns. We used a corpus of all Wikidata discussions and a mixed methods approach with statistical tests, network analysis, and text and graph embedding representations to study members' behaviour and the influence of their communication to the project. We find that Wikidata editors' interactions form a small world network, with the content of a post impacting the continuity of conversations. In the same vein, Wikidata members' age of contributions and their conversations explain long-term engagement. We hope our findings help the Wikidata community improve practices to increase contributions and develop tactics to enhance long-term participation.

CCS CONCEPTS

• Information systems → Data mining; • Human-centered computing → Wikis; Social networks; Empirical studies in HCI.

KEYWORDS

Collaborative Knowledge graph, network analysis, graph embeddings, text embeddings

ACM Reference Format:

Anonymous Author(s). 2018. Talking Wikidata: Communication patterns and their impact on community engagement in collaborative knowledge graphs. In *Proceedings of THE WEB CONFERENCE (WWW '24)*. ACM, New York, NY, USA, 10 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 INTRODUCTION

Online communities are groups of people interacting with each other on online platforms. Online peer-production communities are groups of self-organised people who collaboratively interact towards a shared outcome, e.g., Q&A forums like Quora,¹ open-source

¹<https://www.quora.com/>

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WWW '24, May 13–17, 2024, Singapore

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ACM ISBN 978-1-4503-XXXX-X/18/06...\$15.00

<https://doi.org/XXXXXXX.XXXXXXX>

software projects like Stack Overflow,² and knowledge production projects like Wikipedia³ and the largest collaboratively built knowledge graph, Wikidata.⁴ Online peer-production projects rely on their members for sustainability and success [5]. Understanding the collaboration mechanisms behind these communities is important for the community health. Particularly, in collaborative knowledge graph communities, patterns of interactions can indicate engagement levels which is essential for high contributions, and bonding with the community [7, 11, 16]. Although engagement is important for these communities, many projects lack members' long-term engagement [29, 32] and rely on a small group of members for the majority of contributions [25, 27]. It is thus essential to study community behaviour and identify what impacts communication and coordination.

Previous studies in collaborative knowledge graph communities studied contributor's behaviour by investigating emerging roles [25, 29] and lifespan [32] based on their activity logs and semi-structured interviews. While activity logs can add important information on collaboration patterns, they do not give information regarding coordination or sentiment interactions. Previous studies in peer-production emphasised the importance of contributors' communication and showed that discussions have an impact on the quality of results [31], and contributors' performance [3, 10]. Therefore, communication patterns can add a different view in the investigation of collaborative knowledge graph communities, and give insights regarding their impact on contributors' behaviour.

In this study, we investigated members' behaviour in a large-scale collaborative knowledge graphs, particularly in Wikidata, through members' communication patterns and explored whether these patterns impact the project. Wikidata is an open generic interest knowledge graph supporting numerous intelligent systems [41] with a community of volunteers with diverse backgrounds who collaborate online. We use a mixed-method approach with descriptive statistics, statistical tests, network analysis, as well as graph and text embeddings to understand community interactions. We analysed the behaviour of Wikidata contributors by answering *RQ1 - What are the characteristics of editors' collaborations?*. To study the impact of contributors' communication on the project, we answered *RQ2 - What impacts the course of discussions?*, i.e., given a set of features for a piece of text, what features influence whether it will receive a response; and *RQ3 - Do discussions impact editors' engagement?*, i.e., given a set of features for editors, what influences the probability to be active or dropout from the community.

Our contributions are twofold: (i) a dataset of separated posts from the Wikidata discussions and (ii) a framework to investigate what influences conversations and editors' engagement. Our study shows that Wikidata editor interactions form a small world network with a high clustering coefficient and low shortest path. For

²<https://stackoverflow.com/>

³<https://www.wikipedia.org/>

⁴https://www.wikidata.org/wiki/Wikidata:Main_Page

this interaction network, network topology and the content of discussions influence whether a conversation will continue. At the same time, the age of an editor in the community and their discussions impact their engagement. Our findings can support Wikidata to improve practices and tools. Furthermore, our results can help recommendation systems [1, 45] with improved suggestions for editors' engagement.

2 BACKGROUND: THE WIKIDATA KNOWLEDGE CONSTRUCTION PROJECT

The Knowledge Graph. The term knowledge graph describes a knowledge base that is built and organised as a graph [15] of nodes (entities like *people* and *cities*) and edges (relations like *friend of* or *capital of*). Wikidata is free and openly available for people to use and contribute [42]. As part of the Wikimedia Foundation, it supports other projects like Wikipedia, Wikisource etc. Today, Wikidata includes more than 100M nodes called *items*, 10K edges called *properties*, and a community with more than 24K contributors referred to as *editors*. Items and properties are organised in taxonomies using the properties *instance of* and *subclass of*, similar to *rdf:type* and *rdfs:subClassOf* respectively.

The Contributors. Editors can be either *humans* or *bots*, i.e., software executing repetitive edits. Humans can be registered community members or may choose to contribute anonymously. Registered members can gain additional editing rights, based on their experience with the project (see Appendix A.1).⁵ For example, *Administrators* access level is a group of editors with high administrative and technical rights elected by a community vote. Among others, they can grant rights to others, block spammers and prevent vandalism. Bots are responsible for repetitive work like modification of items or property constraints violations, and their services should be approved by the community. Wikidata has in total 399 bots while approximately 100 are active every month.⁶

The Contributions. People contribute to the knowledge graph in various ways. They edit by creating, revising, or removing content, and asking, answering, or deciding how to structure information. Wikidata is built using a wiki, i.e., a web platform that allows people to edit the project using a web page in the browser [43]. Each item and property has its own wiki page, which progresses over time. All history of edits is publicly accessed. The editing activities, referred to as *revisions*, include metadata for the log such as the page name, the name of the editor responsible for the revision, and a timestamp.

The Communication. Similar to collaborative ontology engineering projects [33], discussions in Wikidata are a tool for coordination and collaboration. Wikidata has two types of *discussions channels*: the talk pages, and the *communication* pages.⁷ Talk pages come with every content page, e.g., item, property, or help page. Editors use talk pages to coordinate activities on the content page, flag mistakes, and raise questions. In addition, in Wikidata, we find various communication pages for discussion and making decisions regarding general concerns, and technical or management support. Examples of communication pages are the *Bot request* page

where people specify the need for bot development or the *Property proposal* and *Properties for deletion* pages for graph construction. Figure 1 shows an excerpt from a Wikidata discussion page. Both talk and communication pages follow a similar structure, with titles referred to as *subjects* separating the raised issues and a sequence of responses referred to as *posts*. Every post should be ended with the contributors' username or IP in the case of anonymous contribution, and a timestamp. A set of posts under a subject title is called *thread*.

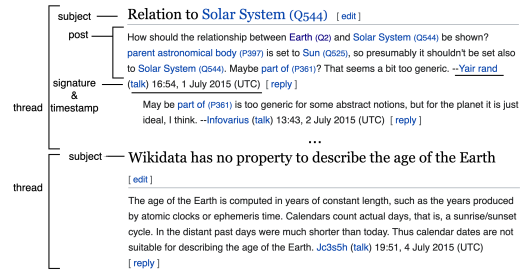


Figure 1: An example of a Wikidata thread.

3 RELATED WORK

3.1 Members' behaviour in online communities

Previous studies on social media, online forums, and peer-production communities, investigated their communication patterns. Several studies focused on antisocial behaviour like anomaly detection (e.g., rumour, spamming, fraud, and cybercrime detection) [22, 30], sentiment detection (e.g., irony or sarcasm, and controversy detection) [13, 17], and manipulation (e.g. sockpuppets) [20, 34] to maintain healthy communities. Furthermore, other studies analysed the structure of communication networks. For example, before the advent of social networks (2006), instant messaging platforms like MSN engaged 245M members in 30B conversations. An analysis of the MSN members network [21] –in which nodes represent members, and edges connect two members if they ever exchanged messages– found characteristics like a power law in degree distribution; high clustering coefficient, $C = 0.1$; and one connected component including $> 99\%$ of all nodes. These are similar to characteristics of a small-world networks [44] (i.e. networks in which node distance L grows proportionally to the logarithm of the number of nodes N) rather than those of random networks [8]. While these analyses are useful to understand the topology of collaborative communities, they do not say much about specific interactions, nor how they affect (e.g. in terms of engagement) members and their work.

3.2 The impact of discussions in peer-production communities

Prior work investigated members' interactions from two views: their editing interactions (i.e. co-production, for example, articles in Wikipedia) [4, 37, 38]; and their communication interactions (i.e. online discussions for work management, cooperation and collaboration) [9, 10]. Previous work in communication interactions explored the impact of discussions on contributors' performance

⁵https://www.wikidata.org/wiki/Wikidata:User_access_levels

⁶<https://stats.wikimedia.org/#/metrics/wikidata.org>

⁷https://www.wikidata.org/wiki/Wikidata:Community_portal

investigating conflicts in open source software communities [10] or Wikipedia [2, 3]. Filipova et al. [10] showed that different types of conflicts, i.e. task, normative, process, etc. affect the results differently. Normative conflicts (i.e. group function conflicts) negatively impacted the intention to remain in the project, and process conflicts (i.e. disagreeing on how to perform tasks) harmed contributors' performance. Besides conflicts, there are other factors, e.g., the absence of an answer to question, that can impact participation and performance.

3.3 Members' behaviour in Wikidata

Previous studies used the activity logs and semi-structured interviews to analyse participation profiles [25, 29, 32] and engagement [27, 28, 46]. Studies classified the editors based on their volume of edits, type of editing activities, and lifespan to analyse their behaviour. Muller et al. [25] found that most editors made specialised contributions, and only a small piece contributed widely to the project. Similar results found by Piao et al. [27], who showed that 0.08% editors contribute 80% of all edits. Piscopo et al. [29] clustered editors based on the volume of edits showing that contributors, performed a lower number of edits and participate less in community discussions, while leaders performed quality control activities and worked on knowledge engineering tasks. Satasua et al. [32] classified the editors based on their volume of edits over time, arguing that long lifespans and high volumes of edits have the highest impact on the project.

Piao et al. [27] aimed to understand what influences contributors' dropouts, using the editors' activity logs. The authors used a variety of statistical and pattern-based features like the number of edits, activity types, duration between edits, pair of entities editing and their characteristics to show the most influential feature is the number of last month's total edits. This is similar to findings in Wikipedia [12].

Previous work in Wikidata discussions used qualitative analysis to understand the content of discussions [18, 19]. Kanke et al [18] analysed Wikiprojects (i.e., sub-group of editors working on specific topics) to find they include coordination like main activities and support like share of rules and tools. Koutsiana et al. [19] analysed talk pages showing that the main topics of discussions are related to knowledge engineering activities. However, contributors' interactions in terms of behavioural patterns were out of these studies' scope.

4 DATA

Wikidata supports publicly available monthly updated dumps with the knowledge graph in JSON and RDF format, as well as, with the history of all edits in XML format.⁸ In our work, we used the XML files updated by February 2023 with all activity logs and discussions. Appendix A.2 presents the description of how we process the raw discussion files to create our dataset with separated threads and posts.

4.1 Data description

We created two dataset from the XML files: a *discussion* dataset (text, username, timestamp, subject) and a *history of edits* dataset

⁸https://www.wikidata.org/wiki/Wikidata:Database_download

Table 1: Descriptive statistics for the editors' categories.

Category	#editors	#posts	#revisions
All editors found in discussions	31,196	1,397,565	-
Registered	21,995	1,012,483	750,126,203
Unregistered	9,130	62,976	-
bots	71	322,106	-

with revisions (username, timestamp) (see Table 1 for descriptive statistics). Using the two datasets, a MediaWiki API,⁹ and network analysis, we created a list of features to explore registered editors' characteristics. Table 2 shows the features and their descriptions.

Regarding label activity status, we built a notion about who is considered an active or inactive editor. Wikidata classifies editors as active and inactive based on their monthly editing activities without considering editors permanently leave. Sarasua et al. [32] highlighted that editors present varying periods of absence and no edits, leading to comeback or permanent leaving. The authors measured these gaps of inactivity and, looking at the percentiles, defined that after approximately 10 months of no contributions, editors are considered permanently inactive. Similar to Sarasua et al. [32], we used the percentile for the maximum gaps of inactivity of registered editors discussed in Wikidata. Based on Figure 2 we chose 85% percentile defining a threshold of 23 months. We found that 67% of editors were active and 33% inactive. It is worth noting that during the 10 years of Wikidata, we found several editors absent for 5 to 9 years but returned to contributing again (see Figure 2). The maximum gap of inactivity was 118 months. These findings differ from the first 4 years of Wikidata, where Sarasua et al. [32] found the maximum gap of inactivity being 16 months and 55% of editors inactive.

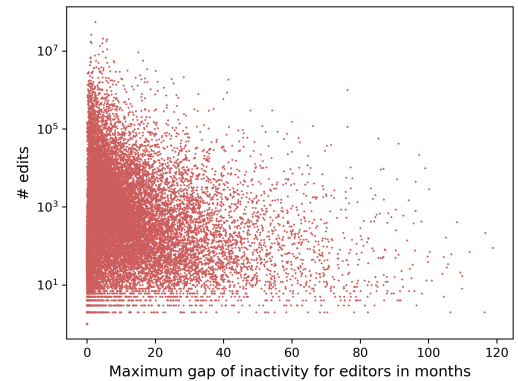


Figure 2: The maximum gap of inactivity in months and the number of edits for registered editors discussing in Wikidata.

5 METHODOLOGY

To answer RQ1 - What are the characteristics of editors' collaborations?, we used network analysis and descriptive statistics. We built and explored three discussion networks: (i) with all editors found in the discussions dataset (NF); (ii) with human editors,

⁹https://www.mediawiki.org/wiki/API:Main_page

Table 2: The lists of features used to investigate editors' characteristics and their descriptions.

Feature	Description
account age	The range in months between the editor's registration date and their last edit timestamp.
# of edits	The total number of edits excluding discussion page edits.
# of posts	The total number of posts in discussion pages.
talking type	Label related to the way editors talk: asking for editors mostly starting threads; responding for editors mostly responding into threads.
activity status	Label related editors' current activity: active for editors with range between their last edit and today (1/3/2023), less than 23 months; inactive for editors with the same range more than 23 months.
access level	The role/roles of editors based on Wikidata hierarchy (Basic level users, Rollbackers, Translation administrators, Administrators, IP block exceptions, Property creators, Confirmed users, Interface administrators, CheckUsers, Wikidata staff, Bureaucrats, Suppressors, Flooders)
degree	The number of edges for a node.
eigenvector	It defines the centrality of a node as proportional to its neighbours' importance. For example, for two nodes in a network with the same degree, the one connected to more nodes with a high eigenvector score should have a higher eigenvector score.
closeness	It counts the average of shortest paths. Nodes with a high closeness score have the shortest distances to all other nodes.
betweenness	It is based on the shortest path between nodes. Counts the number of times a node acts like a bridge.
clustering coefficient	It quantifies how close its neighbours are to be a clique (complete graph).

registered and unregistered (NH); and (iii) with registered editors (NR). Bots participating in discussions is a common practice similar to Wikipedia [23]. Bot controllers participate in discussions or bots deliver automatic messages. We explain more in Section 6.1.2. For the three networks, nodes were editors, humans or bots, who had at least one post on a Wikidata discussion page, and edges connected editors participating in the same thread. We used centrality metrics (see Table 2) [6] to compare the three networks.

We further investigated the NR discussion network, including editors' features. We explored the network's connected components, i.e. connected sub-network that is not part of any other larger network connected component, and the case of Wikidata editor interactions forming a small world network. If we consider a network as connected "friends", in a small world network, not all nodes are friends of another, but the friends of any given node are likely to be friends with each other. A network forming a giant component indicates a small world network.

Small world networks are characterised by a high average clustering coefficient and a low average shortest path [44]. To test our small world hypothesis, we compare these metrics with a similar small world network generated with the Watts-Strogatz method [44], as well as with a similar random network. Finally, we compared our results with the study by Lescovec and Horvitz [21], who studied the MSM network.

To answer **RQ2 - What impacts the course of discussions?** we used a set of features based on editors' characteristics, and graph and text embeddings. We used a hybrid model combining the different features to investigate what influences the course of a discussion in Wikidata, i.e., given a text, can we predict how likely someone will respond?

**Figure 3: Representation of the network for RQ2 and RQ3.**

For *editors' features* (EF), we tested account age (EFage), # of edits (EFedits), # of posts (EFposts), and access level (EFrights) from Table 2.

For *graph embeddings* (G) we built a network including three types of nodes: editors, posts, and thread titles. Editors are connected with their posts, while posts are connected with the thread

they are included (see Figure 3). After testing a number of models like RotatE and TransR, we used CompGCN with embedding dimensions equal to 100 to create graph embeddings for each entity in our network. Due to the structure of our graph and the low number of relations, CompGCN showed the best performance with Hits@1 0.37, Hits@3: 0.56, Hits@5: 0.64, Hits@10: 0.75 and Mean Reciprocal Rank 0.5. For this prediction we used the post embeddings (Gpost)

For *text representations* (T) we used the *text – embedding – ada – 002*, a *GPT3.5* large language model from openAI. At the time of this study, *GPT3.5* was a state-of-the-art model processing long documents [26]. We generated text embeddings for the posts in two ways: given as input the posts individually or part of the thread, i.e., from first to *i* post. Then we tested the probability of a post receiving a response with three cases: (i) the first posts of every thread (TfirstPost); (ii) all posts (Tpost); and (iii) the thread (Tthread). Before generating text embeddings, we preprocessed the raw data for cleaning. Cleaning involved removing signatures and timestamps at the end of the post, replacing the item and property alphanumeric codes, i.e., Q# and P# with [ITEM] and [PROP], as well as any code elements with [URL] or [CODE], and removing any remaining unnecessary punctuation.

We used a neural network for an ablation study, i.e., test each of the above features individually as well as in combinations. We split our dataset to train, test, and validate sets of 60%, 20%, and 20%, respectively. We tested one to three layers with 8, 32, and 64. Section 6 presents the best results from the different combinations. We used "relu" and "sigmoid" for activation, "ada" for optimization, and "binary_crossentropy" for loss. We applied a weighting layer and a cutoff for minimum loss. We evaluated results using precision, recall, accuracy, and F1.

The large network size, including more than 2.5M, required a high computational power for the graph representations. For this reason, we used a sample of our data to answer RQ2 and RQ3. We used threads found in item talk pages. Compared to all discussion channels, item talk pages presented the best balance between the posts receiving a response and those not (44% of posts had a response, 56% did not). Furthermore, prior work shows that in item talk pages, 50% of posts starting a new thread do not get attention from the community to receive an answer [19]. Our study may help reveal factors that influence responses that can support the project to improve communication.

To answer **RQ3 - Do discussions impact the editors' engagement?**, we used a similar strategy and the same sample data with *RQ3* to explore what influences editors' engagement. We tested the same editors' features, graph embeddings for the editor nodes (Geditor), and text embeddings for their posts. For each editor, to generate their text embeddings, we took the average of all their post text embeddings (Tavr). We defined editors' engagement using the feature status as described in Section 4.1, i.e. the possibility an editor be *active* or *inactive*. Our sample included 75% active and 25% inactive editors.

Finally, to further investigate our findings and the statistical significance between the features, we use a variety of statistical tests based on the case. Our code and data are publicly available.

6 RESULTS

6.1 RQ1 - What are the characteristics of editors' collaborations?

6.1.1 Summary. We found that a quarter of registered editors are disconnected from the network, meaning they did not receive responses to their posts, with half of them being inactive today. In contrast, bots were the most connected editors. In addition, we showed that the Wikidata interaction network forms a small world network and a random node can reach 90% of the nodes in 4 steps. We found that 20% of editors are actively discussing and making decisions, with most of them being members for more than 6 years. We also found that the majority of editors discuss are *basic level users*, with some of them presenting the highest number of posts.

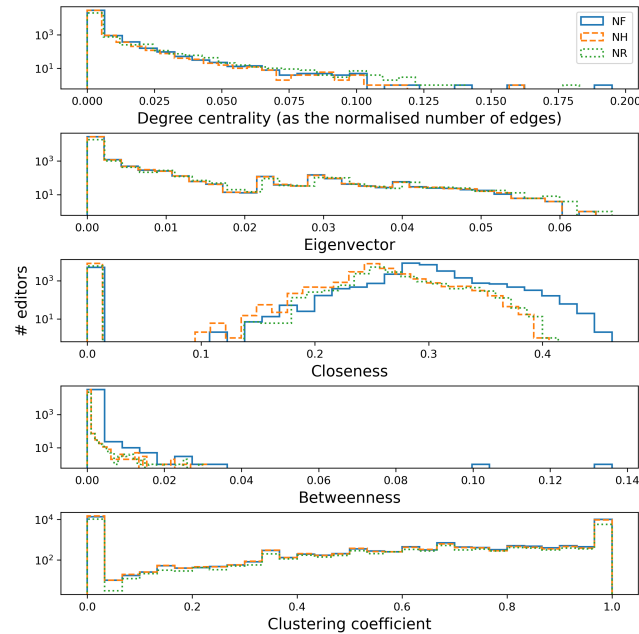


Figure 4: Centrality metrics for the three networks

6.1.2 Comparison of the three discussion networks. Figure 4 presents centrality metrics for the NF, NH, and NR networks. For all networks, degree centrality, i.e. “how many friends I have”, presents a

high number of editors, 14% NF, 24% NH, 26% NR, with no edges. This indicates that a quarter of human editors did not receive responses when starting a discussion. Furthermore, 47% of them are inactive, i.e., half of the editors asked but not receive a response dropout, and the rest continued contributions without discussing. We further analysed the relation between the feature *answer of a post* and the *status* of the editor who wrote it using the Chi-square test for categorical variables [39]. We found a p-value less than 0.05, which means the two features are related.

For eigenvector and clustering coefficient metrics, the three networks presented similar distributions (see Figure 4). The eigenvector shows “how well-connected are my friends” and the clustering coefficient shows “if a friend of a friend is my friend”. We did not find nodes with better or worse connected friends nor clustering for registered, unregistered and bots.

For closeness, we found higher values for NF, but NH and NR present similar distributions. Closeness shows “the distance I have with my friends” with high closeness for nodes with short distances from the other nodes (see Figure 4). At the same time, betweenness, i.e. “how many times I connect friends”, presented lower values for NF. The higher values for closeness and low values for betweenness for NF indicated the importance of bots in the network, which bridge distances between nodes and make paths for all nodes shorter. Bots are the most connected editors in the network with two of them presenting with great difference the highest betweenness.

Following the above analysis, we investigated which nodes presented the highest centrality metrics for the three networks. For the NF network, a bot had the highest degree, closeness and betweenness values. “DeltaBot” was the most active editor in discussions, responding to requests for deletion or merging of items after checking their status with prefixed answers such as “on hold This item is linked from 3 others”. For the NH and NR networks, editors with administrative access levels, like Property creators or Administrators, received the highest metrics. This indicated the importance of bots in communication related to maintenance jobs and the commitment of the editors with higher level access to communication.

6.1.3 Investigation of the network with registered editors. We studied the hypothesis that Wikidata editors' interactions form a small world network for the NR. We first explored connected components of the NR. We found 5,722 connected component with the largest component including 73% of the nodes (16,169 nodes and 470,126 edges) and the second largest only 0.04%. This indicated that NR comprises a giant component and many very small components. This aligns with the above findings that 26% of nodes did not present any connection.

We then measured and analysed the average cluster coefficient and the average shortest path of NR. We found that the average clustering coefficient for the NR is 0.6 and the average shortest path is 3. A similar small world network has an average clustering coefficient of 0.1 and an average shortest path of 3. Furthermore, a random network with similar characteristics has an average clustering coefficient 0.004 and an average shortest path 3.

The giant component, the high average clustering coefficient and the low average shortest path are characteristics of a small world network. NR presented a very high clustering coefficient meaning that in many cases, the nodes' clustering coefficients are

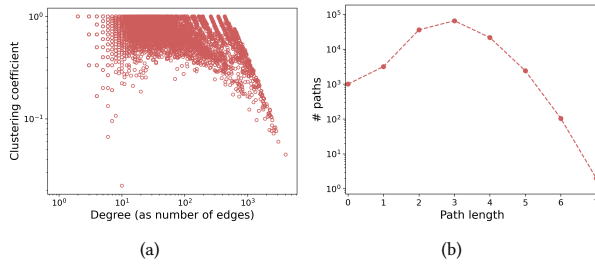


Figure 5: (a) Clustering coefficient (b) Shortest paths

equal or close to 1. We see this in Figure 5 (a) even for nodes with high degrees. Previous results on MSN network found a clustering coefficient equal to 0.1. This suggests that clustering in the NR network was 6 times higher than the MSN, forming a much clustering network.

NR presented a very low average shortest path with a random pair of nodes on average 3 steps apart. Similar to the MSN study [21], we randomly sampled 1000 nodes and calculated their shortest paths to all nodes. Figure 5 (b) shows the distribution of shortest paths. The longest shortest path is 7 showing that long path do not exist in NR. This is different from the MSN network where a random pair of nodes can be reached on average of 7 steps, and there are long paths up to 29 steps [21]. Looking at the percentile of shortest paths, we found that for the NR one can reach 90% of nodes in 4 steps, while for the MSN in 7.

The high clustering coefficient and the low average shortest path with the absence of long distances of NR are characteristics of a small-world networks suggesting that has many clusters, and many nodes in these clusters are connected with other clusters.

6.1.4 Investigation of registered editors characteristics. Figure 6 shows the number of edits, posts and account age distribution for the registered editors. For the number of edits and posts, we found a power law distribution. Most editors, 80%, presented less than 5K edits and 10 posts. This indicates that most participants contributed with low edits and did not engage in discussions. This result suggests that the decisions made within the Wikidata community are taken from the 20% of the editors. For the distribution of the account age, we found 24% of editors stay in the community for less than a year and 10% for editors staying engaged in the project for more than 10 years. This result is similar to the first four years of Wikidata where many users have a short-lasting account age and a few longer-lasting accounts [32].

Figure 7 shows the number of posts to account age for the different access roles we found discussing. In the figure, we see that editors with more than 6 years in Wikidata were the most active in discussions, with the most active a *property creators* followed by *basic level users*. Furthermore, *basic level users*, with great difference, is the access level participating the most in discussions (see Figure 8). Next, we explored the editors' access level (see Figure 8) for *askers*, i.e. editors usually started a thread, or *responders*, i.e., editors usually participated in threads (see Appendix A.1 for access level description). The majority of editors were *basic level users* followed by *rollbackers*, *translator administrators*, and *administrators*.

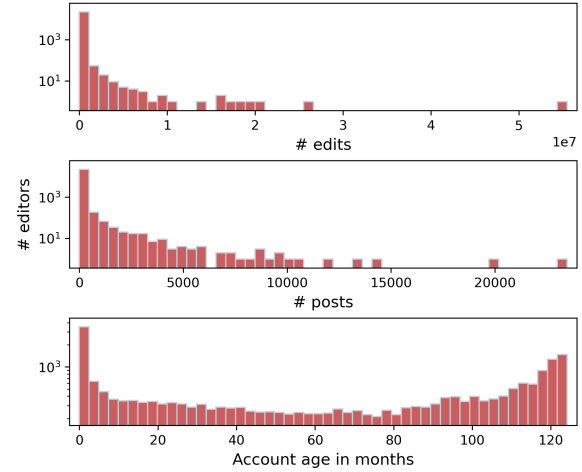


Figure 6: Frequency of editors to their characteristics.

In their majority *basic level users* and *rollbackers* were askers. In contrast, administrative roles like the different types of *administrators* and *bureaucrats*, as well as knowledge graph experts like *property creators*, are responders. This was expected if we consider that *basic level users* seek advice or request access and tool support from the community and particularly from members with higher access levels.

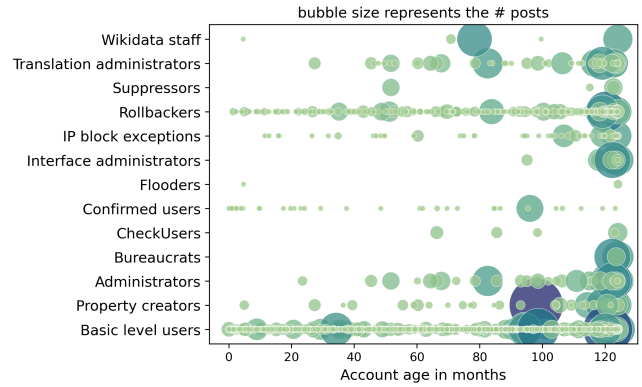


Figure 7: Number of posts and the account age for the different access levels.

6.2 RQ2 - What impacts the course of discussion?

6.2.1 Summary. We found the highest performance for graph embeddings and text embeddings. For text embeddings, we showed small differences between the type of text we give to the model, i.e. one post or thread. In contrast, the editors' characteristics did not perform well.

6.2.2 Prediction results. Graph embeddings presented the highest results between the individual features for the prediction of a post

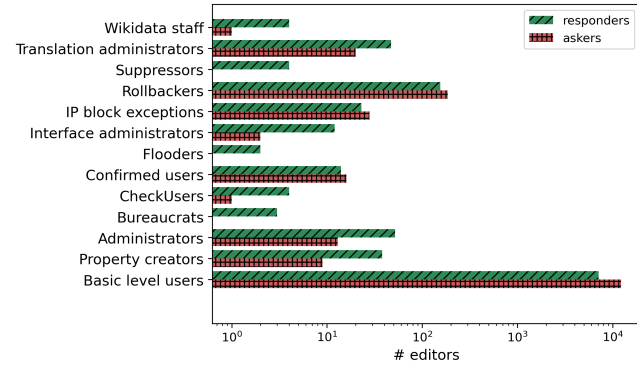


Figure 8: The histogram presents the number of editors and the talking type for the different access levels.

Table 3: Results for the model predicting whether a post will receive a response.

Features	Precision	Recall	Accuracy	F1
EFage	0.54	0.55	0.55	0.54
EFedits	0.56	0.57	0.57	0.56
EFposts	0.59	0.58	0.58	0.57
EFrights	0.55	0.56	0.56	0.53
EF	0.59	0.59	0.59	0.59
Gpost	0.72	0.72	0.72	0.72
TfirstPost	0.7	0.7	0.7	0.7
Tpost	0.68	0.68	0.68	0.68
Tthread	0.69	0.69	0.69	0.69
EF + Gpost	0.73	0.72	0.72	0.72
EF + TfirstPost	0.7	0.69	0.69	0.7
Gpost + TfirstPost	0.8	0.79	0.79	0.79
Gpost + Tthread	0.76	0.76	0.76	0.76
EF + Gpost + TfirstPost	0.78	0.78	0.78	0.78

to receive a response. This is expected if we think of the topology of the network (see Figure 3). A post connected with a thread with a high degree, i.e., the thread is connected with a high number of posts, will have a higher probability of receiving a response. In addition, we found high performance for the text embeddings with little difference between the TfirstPost, Tpost, and Tthread. This indicates that the content of the text had an influence in getting a response. This is essential for the TfirstPost results, which mark the beginning of a discussion. The content of a post impacts whether the thread will get a response and continue the conversation.

The best indication for a post to receive a response was the combination of its link topology and its content (Gpost + TfirstPost). In contrast, editor characteristics did not show considerable influence on the prediction. This suggests that it did not matter who wrote the post, e.g., if it was a long-lasting account with administrative access levels, but where it was written and what it said.

6.3 RQ3 - Do discussions impact the editor's engagement?

6.3.1 Summary. We found the highest performance for the account age and text embeddings. Similar to RQ3, the rest of the editors' characteristics, particularly access level, did not perform well.

Table 4: Results for the model predicting the editors' status.

Features	Precision	Recall	Accuracy	F1
EFage	0.83	0.72	0.72	0.74
EFedits	0.74	0.54	0.54	0.55
EFposts	0.71	0.57	0.57	0.59
EFrights	0.76	0.31	0.31	0.21
EF	0.83	0.73	0.73	0.75
Geditor	0.63	0.63	0.63	0.63
Tavr	0.66	0.69	0.69	0.67
EF + Geditor	0.67	0.66	0.66	0.66
EF + Tavr	0.78	0.75	0.75	0.76
Geditor + Tavr	0.65	0.65	0.65	0.65
EF + Geditor + Tavr	0.69	0.7	0.7	0.69
EFage + Tavr	0.78	0.75	0.75	0.76

6.3.2 Prediction results. The account age shows the highest performance between the features for the editors' status. This may indicate that the longest an editor participated in Wikidata, the most likely to be active. Looking at the status of editors with more than 6 years of account age, we found that 85% of them are active. To further investigate this assumption, we tested the statistical difference between the *account age* and the *status* of editors using the Mann-Whitney U non-parametric test [24]. We found a p-value equal to zero, which means that all the values in one class, i.e., the values of age account for the active editors, are greater than all the values in the second class, i.e., of inactive editors.

It is worth noting that the rest of editors' features, and specifically EFrights, presented very low influence. This indicates that an editor with higher rights was not more likely to be active than a *basic level user*. We found that editors' text embeddings also showed an influence on their status. This text includes all the posts that an editor wrote. This result may suggest that the sentiment and the topic of discussions impacted their engagement. The best performance was the combination of their account age and the content of their posts.

7 DISCUSSION

We analysed the Wikidata discussion by studying the structure of editors interactions network, and exploring what influences the continuity of a discussion as well as editors' engagement. Our findings highlight the value of investigating communication interactions to understand what influences engagement and support collaborative communities to improve practice and keep and increase the number of members and their contributions.

Similar to previous studies in Wikidata [19], we highlight the importance of posts that did not receive a response. We found a relation between posts with no response and the editors' status, with 47% of editors who did not receive a response to their posts dropping out, and the rest did not attempt to participate in any discussion in the future. Prior work suggested that reasons for no responses could be the numerous item talk pages leaving posts unattended and the lack of guidance for using talk pages compared to Wikipedia. We found that the content of posts is also an influential factor. Koutsian et al.[19] explored topics of item discussions presenting that 46% of topics are related to knowledge engineering. The editors discuss content page management, suggest or report changes, and ask for help and specifications. The complexity of the subject, particularly for a group of people with diverse knowledge

engineering experience, may make it difficult for editors to express the subjects in a simple and informative way. This may prevent responses to obscure posts. This results suggest that guidance in asking questions and discussing changes may increase the possibility of continuing a conversation. Wikidata use templates for specific discussion channels like *Request for a comment* and (Property proposal) to help the community receive all the information needed to make a decision on a subject. In item talk pages, there is no proposed template to start a discussion. Using suggested practices to write a post may enhance communication and help, especially new members, to receive answers. This could increase new members' participation. Previous research on GitHub [14] has emphasised the importance of guidance in discussions for members' familiarity and tool support for effective discussions.

In the same vein, our findings suggest that editors' engagement is influenced by their account age and the content of their discussions. We showed that the longer the account age, the more likely to be active. This indicates that one important factor for engagement is the first period an editor contributes to the project. In addition, the content of editors' posts can also influence engagement. In a study related to open source software [35], the authors find that 30% of developers receiving nonacceptance in their pull-request are disappointed and hesitate to place another pull-request. Moreover, prior work on the impact of conflicts indicates that conflicts negatively impact the intention to remain in the project [10]. In the Wikidata scenario, Koutsiana et al. [19] showed that Wikidata item talk pages had low frequency in conflicts [19]. This can indicate that editors did not receive aggressive or confrontational feedback leading to dropout, but the posts' character, e.g. most posts are questions, may influenced the prediction. As mentioned earlier, in the same study, the authors argued that the topics discussed mainly relate to knowledge engineering activities. The complexity of knowledge engineering, especially for members with no prior experience in building knowledge graphs, with questions receiving insufficient or confusing feedback could lead to disappointment and resignation. These results highlight again the importance of first experiences for engagement when new members join the community. The community may need to monitor whether new members ask for help, developing practices for guidance and support to encourage them so as to overcome a period of time participating to stay engaged.

We find that Wikidata discussion interactions form a small world network with high average clustering coefficient and low average shortest path. For Wikidata, this result can be explained if we consider the discussions classified into the different discussion channels based on their topic, i.e. *Property proposal*, *item talk pages*, etc.. Editors discussing in specific discussion channels are likely crossed by threads. However, our results indicate that many members are not limited to one kind of discussion channel. The small world characteristics facilitate the propagation of information much faster than random networks [21, 36]. Wikidata could benefit from this for tool adoption, spam detection or vandalism. However, this quick spread of information could potentially be harmful in case of deliberate spread of "fake news", e.g. face practices.

7.1 Limitations

For this study, we used a sample of discussion and only quantitative analysis to investigate the impact of text to the course of discussion and editors' engagement. We analysed item talk pages, but including different types of discussion channels may provide extra details to the prediction models. Furthermore, qualitative analysis similar to previous studies about the content discussed [18, 19] could reveal insights for the influential topics. In addition, we tested one network structure for the predictions. Experimenting with different designs may improve results for the post and editors' topology, giving another understanding of these predictions.

8 CONCLUSION

We investigated the Wikidata editors' interactions through their lens of discussions. We used a mixed method approach using descriptive statistics, network analysis, and graph and text representations. We found that the Wikidata community forms a small world network with news propagating fast across the network. At the same time, we suggested that the longer an editor stays in the community and feels inclusive, the more likely to stay engaged in the project. We hope our findings can support Wikidata in developing new practices to help new members find their steps in the project to stay engaged, as well as help project designers transmit new tactics to the community. Future work could investigate the content of the text and justify what makes a post a good candidate to receive an answer and what discourages members from responding to others. In the same vein, it would be interesting to investigate the text that makes editors stop contributing. This will give an in-depth understanding and enable Wikidata to improve practices and tools.

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A APPENDIX

A.1 Access levels

A list of access levels as defined by Wikidata and their descriptions. We included only the access levels of editors we found in discussions.

Table 5: List of access levels and their descriptions (source).

Access level	Description
Basic level users	Editors with basic editing rights.
Confirmed users	Editors accounts with at least 4 days of age and at least 50 edits become autoconfirmed. Editors which do not meet the autoconfirmed criteria can be granted the confirmed.
Property creators	New properties can only be created by users in this group or administrators, as properties should be discussed before being used.
Rollbackers	Rollback is a simple tool that can allow fast reverting of vandalism or other obvious abuse. Rollbackers can control this tool
IP block exempt users	If an IP address is blocked with the option to not allow logged-in users to edit, this permission allows editing from such IP addresses
Administrators	These are trusted editors with tasks such as blocking. They can assign rollback, confirmed user, IP block exempt, autopatrolled, and property creator status to other users.
Interface administrators	Editors with the ability to edit sitewide CSS/JS pages and change how content is styled, change the behavior of pages.
Translator administrators	Because Wikidata is a multilingual wiki, it uses the translate extension and translation administrators are users with access to administrative functions of the extension
Bureaucrats	They can grant or remove rights for administrators, bureaucrats, bots, flooders, and translator administrators.
CcheckUsers	These editors examine the IP address data of logged-in users, which is otherwise private information. Its primary purpose is to investigate sock puppets.
Suppressors	Suppression refers to hiding revisions, usernames in edit histories and logs, or portions of individual log entries. Suppressors can have access to these data.
Flooders	Users may be granted a “flood flag” and be flooders when doing repetitive, non-controversial edits. This is to distinguish those actions from vandalism.
Wikidata staff	To facilitate the development of Wikidata, some Wikimedia Deutschland staff members technically hold full administrator access.

A.2 Data preprocessing

During the data preparation, we processed raw text data. Figure 9 shows an example of raw text from a discussion page. In the figure, we can see that raw text follows a template for every page using specific punctuation to identify the subjects of the threads (i.e. `=== subject ===`), and the username (i.e., `[[User: username|username]]`) and timestamp (i.e., hour: minutes date month year (UTC)) at the

end of each post. Similar to previous studies in Wikipedia [23, 40], we phased multiple challenges in the process of the raw text discussions. We processed the text taking place on a discussion page, starting with separating threads and posts on the page, and thereafter extracting the information of the contributors who wrote the posts, and when. Due to the large size of the Wikidata archived dumps, 1.6T, the minor changes in templates happened in the project over the years, the differences in terms of content (e.g., several pages included tables or different types of headers for subjects based on page purpose) in the different communication pages, and the human error, we expected a small percentage of false in our dataset when separating threads and posts, as well as in the editors' usernames. Particularly for the usernames, we phased three types of difficulties: some posts missed the expected username signature in the end, probably due to inexperienced with Wikidata practices editors (this is similar to Wikipedia for 8.3% of posts [23]); the multilingual nature of the projects makes it hard to follow the structure of the template in all languages, and track the username in the raw

text; and the freedom giving by Wikidata of creating usernames with colours, highlights, and emojis, made hard to identify the usernames due to the punctuation related to this features we find in the raw text. Our code and dataset are available in supplementary material.

=== Relation to {{Q|544}},Yair rand ===

How should the relationship between {{Q|2}} and {{Q|544}} be shown? {{P|397}} is set to {{Q|525}}, so presumably it shouldn't be set also to {{Q|544}}. Maybe {{P|361}}? That seems a bit too generic. --[[User:Yair rand|Yair rand]] ([[User talk:Yair rand|class=""signature-talk"">{{int:Talkpagelinktext}}]]) 16:54, 1 July 2015 (UTC)

: May be {{P|361}} is too generic for some abstract notions, but for the planet it is just ideal, I think. --[[User:Infovarius|Infovarius]] ([[User talk:Infovarius|class=""signature-talk"">{{int:Talkpagelinktext}}]]) 13:43, 2 July 2015 (UTC)

Figure 9: An example of a raw Wikidata thread from the item “Earth”.